

Cover Letter

To: Guest Editors, IEEE Journal of Selected Topics in Signal Processing

Special Issue: *Autonomous and Evolutive Optimization in Networked AI*

(Lead Guest Editor: Prof. Liang Song; with Profs. J. Liu, A. Dvir, A. Skodras, V. C. M. Leung, and Q. Bi)

Dear Guest Editors,

We are pleased to submit our manuscript, “**Interpretable Federated Kolmogorov–Arnold Surrogates for Autonomous and Evolutive Resource Optimization over Non-Stationary LEO Links**,” for consideration in the Special Issue on *Autonomous and Evolutive Optimization in Networked AI*.

Scope and fit. The Special Issue calls for online, self-evolving mechanisms that let distributed models improve in multi-agent, time-varying environments, with emphasis on the foundations of signal processing for networked AI, networked AI for cognitive communications, and online drift detection and compensation. Our paper addresses exactly this setting. We study autonomous resource optimization over low Earth orbit (LEO) links, where the per-instance optimum has no closed form and the operating regime drifts as satellites pass and interference handovers occur. We cast the problem as learning the *solution map* of an energy-efficiency power-allocation problem, harden it into a mixed-integer program, and decompose it into an inner statistical-learning task and an outer communication-control task whose static optimum we derive in closed form. A federated Kolmogorov–Arnold network learns the solution map online, while a contextual-bandit controller learns, from feedback, the per-slot update sparsity (the operating point implied by the Lagrangian dual price of the uplink budget), unifying supervised learning with online decision making, as the call envisions.

Contributions and findings. Because the energy-efficiency objective is flat near its optimum, the achieved utility regret is comparable across surrogates; our claim is therefore not a raw performance win but that the KAN is the more *trustworthy* surrogate along four axes a black-box multilayer perceptron (MLP) does not jointly provide. Over twenty seeds with paired significance tests: (i) it is more sample-efficient on the solution map, about $1.8\times$ lower normalized error at equal parameters (Wilcoxon $p < 0.01$), while a linear predictor and a model-based deep-unfolding solver are both clearly worse; (ii) it extrapolates to unseen interference regimes roughly an order of magnitude more accurately, where the parameter-matched MLP diverges on a third of the seeds; (iii) its additive-spline structure yields closed-form, auditable design rules a black box cannot provide; and (iv) a learned sparsification controller halves the uplink while remaining competitive. Every component is isolated by one-factor ablations.

Relation to prior work. Existing learn-to-optimize methods for wireless resource management are almost exclusively black-box networks (fully connected, deep-unfolded, or graph-based) evaluated under stationary, centralized conditions. Our work differs on three axes that matter for autonomous deployment: it is interpretable (the learned policy reduces to closed-form rules), it is studied under genuine non-stationarity with an online communication budget, and it quantifies sample efficiency and extrapolation rather than only in-distribution accuracy. To our knowledge this is among the first uses of Kolmogorov–Arnold networks as interpretable optimization surrogates inside a networked, federated signal-processing pipeline.

Rigor and transparency. In keeping with reproducible, trustworthy research, all results are multi-seed with standard deviations and Wilcoxon signed-rank tests, every comparison between model classes is parameter-matched and trained under an identical budget, and the ablations toggle one component at a time so that effects are attributable rather than correlational. Importantly, we report transparently the cases where our method does *not* help: online grid self-evolution gives no significant gain over a well-sized static grid, and under abundant centralized data or large channel-estimation error the advantage narrows. The complete

code, configurations, and result files that reproduce every number, table, and figure in the paper are openly available.

Declarations. This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submission and declare no competing interests. The work was supported in part by the Posts and Telecommunications Institute of Technology (PTIT).

We believe the paper is a strong fit for the Special Issue and hope you find it suitable for peer review. Thank you for your time and consideration.

Sincerely,

Phuc Hao Do (corresponding author)

Center for Advanced Innovation, Research and Application,

Da Nang Architecture University, Da Nang, Vietnam

E-mail: haodp@dau.edu.vn

on behalf of co-authors Huu Phu Le and Truong Duy Dinh